PAL - Primitive Axiom Layers

Spine v1.3: Publish-Ready Structural Edition

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Status: Public preprint and versioned structural framework; not a completed formal theory
Date: July 16, 2026
Prior Zenodo record DOI: 10.5281/zenodo.20966698

Scope: A0-A15 structural spine, plus a typed outward interface to PECAN. PALET, MAPS, PEA, and Seed, Not Feed remain separately authorized parts of the larger open framework.

Give people something to grow from. Do not consume all the growing space.

Recommended citation: Pang, Christopher D. (2026). PAL - Primitive Axiom Layers: Spine v1.3, Publish-Ready Structural Edition. Zenodo version lineage: 10.5281/zenodo.20966698.

Abstract

Primitive Axiom Layers (PAL) is a layer-language for describing how unresolved possibility becomes readable distinction, how distinction becomes persistent trace, and how persistent structures become relational, recursive, addressable, revisable, and capable of declared closure. PAL does not claim to derive physical law, prove a cosmology, answer why existence exists, or generate ethical authority automatically. It supplies a typed structural grammar and a set of explicit burdens for any later mathematical, physical, computational, or ethical realization.

This v1.3 edition preserves the mathematical and cross-framework repairs consolidated in v1.2.5 while preparing the spine for public citation. A6 separates Lyapunov stability under initial-condition perturbation from structural persistence under perturbation of the governing dynamics; KAM-type persistence and Greene-style breakup diagnostics are not presented as the same claim as trajectory-neighborhood stability. The PECAN Authorization Cut carries a declared activation criterion, a prohibition on normative backflow into PAL identity, and a bounded checker-authority ceiling. A13 distinguishes progress-bearing scheduled return from both Poincaré recurrence and Nyquist-Shannon apparent repetition. The sourced-realization crosswalk records local source roles without treating any citation as proof of the full spine. A12 remains intentionally without a claimed external domain realization.

The result is a strict A0-A15 spine with mathematical definitions, model-level axioms, realization notes, explicit authority ceilings, and open burdens. The Riemann Ledger (RL) is treated as a formal-methods laboratory for these structures, not as independent evidence that PAL is true.

1. Authority and notation

1.1 Claim classes

Every substantial statement in this draft belongs to one of six classes:

1. **Primitive declaration:** a starting term whose content is not derived inside PAL.
2. **Model axiom:** a rule imposed on a declared PAL model.
3. **Definition:** notation or a constructed object inside the model.
4. **Internal consequence:** a result following from declared axioms and definitions.
5. **Domain realization:** an independently established mathematical or physical structure that may instantiate part of PAL.
6. **Interpretation or open question:** a proposed reading without theorem-level authority.

No statement changes class merely because similar structure appears in several domains.

1.2 PAL model schema

After the first distinction has become readable, a PAL model may be represented provisionally as

$$P=(X,B,K,R,T,C,\Pi),$$

where:

- X is a typed state or carrier class;
- B is a class of declared boundaries;
- K is a class of trace carriers;
- R is a class of permitted relations and routes;
- T is a family of transport or transformation maps;
- C is a class of certificates, receipts, and unresolved burdens;
- Π is a family of boundary projections from richer carriers to locally readable judgments.

This schema is unavailable at A0 because it already assumes distinctions, types, and relations. It is introduced only as later bookkeeping.

1.3 Core typing rule

For any event e meeting a boundary B,

$$e \neq \iota_B(e) \neq Cert(\iota_B(e)),$$

where e is the occurrence packet, $\iota_B(e)$ is its boundary-relative imprint, and $Cert$ is a claim-bearing certificate. Equality may be asserted only after an explicit equivalence map and its domain have been declared.

1.4 Projection non-identity

A locally readable output is generally a quotient of a richer carrier:

$$j_B=\Pi_B(K),\Pi_B(K)\neq K \text{ in general}.$$

A bit, label, or terminal status therefore does not automatically preserve the full trace that justified it.

1.5 Source-role discipline

A citation may support a definition, theorem, realization, analogy, or open frontier without supporting every claim made near it. For a source s cited in relation to claim C, record

$$Role(s,C)\in\{DEFINITION,THEOREM,REALIZATION,ANALOGY,FRONTIER\}.$$

The source role is local to the cited claim. In particular,

$$THEOREM(s,D) \text{ does not imply } THEOREM(s,\text{PAL as a whole}),$$

even when the independently established structure D is a precise realization of one PAL layer. Sources may strengthen definitions, examples, and auditability; they do not automatically prove the spine's ontological interpretation or its cross-layer bridges.

2. Optional pre-spine territory: A_{-1}

A_{-1} names proposed causes beneath or prior to PAL's formal floor: God, a simulation, a prior universe, necessary being, or another metaphysical source. PAL neither affirms nor denies such proposals. They remain outside the scientific spine unless a readable trace and a valid bridge authorize a more specific claim.

PAL's answer to "Why existence?" is therefore bounded:

> *PAL does not explain why there is something rather than nothing. It explains why that question cannot be asked from nowhere.*

The poetic form may be retained as interpretation, not theorem:

> *Why existence? Because existence turned far enough to leave a trace, persisted long enough to inherit it, and eventually asked.*

A0 — Ω: Unresolved Primitive Possibility

Primitive declaration

Ω names unresolved primitive possibility: the honest formal floor before a readable distinction has authorized objects, sets, measures, logic values, geometry, time, energy, agents, or causes.

PAL uses the symbol Ω while denying that the symbol captures its referent:

$$[\Omega)_P \neq \Omega .$$

The left side is PAL's formal placeholder. The right side is the unresolved primitive that the placeholder does not exhaust.

Axiom A0: unresolved-floor restraint

No later model object may be identified with Ω merely because it is unknown, unobserved, empty, infinite, zero-valued, or outside the present aperture.

If U_B denotes remainder unresolved relative to boundary B, then

$$U_B \subseteq \text{model-relative unresolved material} ,$$

but PAL does not infer

$$U_B = \Omega .$$

Ω supplies formal possibility, not physical energy, computational work, probability mass, or a finite reservoir. Resolution does not permit a conservation equation over Ω.

Stop condition

No theorem inside a later PAL model proves the positive ontology of Ω, because the theorem already operates after distinction.

Primitive θ — Admissibility of Turn and Oriented Readability

θ is not an additional numbered layer. It is the primitive admissibility condition under which a candidate consequence may be recognized as oriented enough to count as a first readable distinction.

θ is not a cause, force, substance, fuel, measurable angle, truth value, or function from Ω. It does not make a distinction occur. It constrains what PAL is permitted to call an A1 distinction after a candidate mark is presented.

The familiar shorthand

$$\Omega \overset{\theta}{\Rightarrow} d_1$$

is retained only as a mnemonic for the layer boundary. It is not ordinary function application or a generative equation: Ω has not been declared a set, no element of Ω is mapped, and no mechanism is asserted beneath the first readable trace.

Minimal admission test

For a proposed A1 witness d_1, let

$$J_\theta(d_1) = J_{asym} \wedge J_{orient} \wedge J_{sep} \wedge J_{minimal}.$$

The four terms require:

1. **Boundary-readable asymmetry:** the candidate presents a readable non-equivalence rather than an undifferentiated placeholder.
2. **Minimal orientation:** the candidate permits a local side distinction without importing geometric direction, angle, or metric.
3. **Typed separation:** occurrence, boundary-oriented imprint, and any certificate about that imprint are not collapsed into one object.
4. **No later import:** the candidate does not smuggle in time, energy, probability, truth valuation, causation, arithmetic primes, persistence, personhood, or authority.

PAL expresses the admission rule in its metalanguage as

$$\frac{J_\theta(d_1)}{Admit_{A1}(d_1)}(\theta).$$

This rule may inspect the proposed A1 packet without claiming that its packet types existed as object-level structure before A1. θ names the constraint on admission, not an entity acting below the formal floor.

Truth-aptness restraint

θ does not supply truth, falsity, reference, a proposition, a valuation, or a semantic domain. At most, oriented readability opens the formal room in which a later addressable statement could become truth-apt.

> *θ does not make a mark true. It makes explicit the minimum under which PAL may admit the mark as readable.*

The first use of “orientation” remains boundary-logical: one side is not the other. Geometric direction, axes, angles, noncommuting rotations, and comparative frames require later layers.

Authority stop

No conforming use of θ may answer why the distinction occurred, supply the work that produced it, or justify any later PAL structure. “θ caused X” exceeds the authority of the primitive.

A1 — First Readable, Orientable Distinction

Definition

An A1 distinction packet is

$$d_1 = (B_1, \sigma_1, a_1, U_1),$$

where B_1 is the first readable boundary, σ_1 is a minimal side-orientation, a_1 is a provisional address, and U_1 is everything not resolved by that distinction.

The boundary-relative sign map may be written

$$\sigma_1 : N(B_1) \to \{-1, 0, +1\},$$

with 0 reserved for the boundary/contact class and ± 1 for opposed sides when such a local neighborhood is meaningful. This sign structure is a later formal representation of A1, not a claim that physical integers existed before distinction.

Axiom A1: co-generation of sides and boundary

A readable distinction jointly establishes a boundary relation and distinguishable sides. PAL does not assume a pre-existing geometric plane that was subsequently cut.

An unsigned occurrence trace and an oriented imprint must remain separate:

$$o(e) = +1, \iota_{B_1}(e) = (-1, +1)_{B_1}.$$

The first expression records that an event occurred. The second records how a boundary oriented its readable consequence.

Time restraint

A1 permits succession to become possible but does not by itself supply duration, a clock, or a before/after comparison. A lone mark has no inherited second state against which to be ordered.

A2 — Retention, Trace, and Seed-Persistence

Definition of a trace carrier

A provisional carrier at indexed moment t is

$$K_t = \left(v_t, \Theta_t, T_t, Cert_t, U_t\right),$$

where:

- v_t is a domain-appropriate irreducible-axis or normal-form address;
- Θ_t is accumulated, preferably unwrapped, orientation;
- T_t is inherited table or transformation authority;
- $Cert_t$ is provenance and proof-of-lineage;
- U_t is unresolved carry.

Ordinary arithmetic primes are not primitive at A2. They are one later realization of a broader irreducible-address requirement.

Axiom A2: retained readability

A distinction persists when enough of its imprint remains readable to constrain or inform a later boundary transition.

For two indexed states, persistence is certified by a permitted transport map

$$\Phi_{t \to t'} : K_t \to K_{t'}$$

and an invariant family I satisfying

$$d_I\left(I\left(K_{t'}\right), I\left(\Phi_{t \to t'} K_t\right)\right) \le \varepsilon_{str},$$

where ε_{str} is a declared structural tolerance. Exact identity is the special case $\varepsilon_{str} = 0$.

Persistence does not require invariant numerical value. A boundary-indexed family may persist through lawful transformation:

$$X_{\lambda_1} \xrightarrow{\Phi_{\lambda_1 \to \lambda_2}} X_{\lambda_2},$$

even when $X_{\lambda_1} \neq X_{\lambda_2}$.

Certificate stripping

If an exterior certificate is removed, the imprint may still survive:

$$StripCert(K_t) = (v_t, \Theta_t, T_t, \varnothing, U_t).$$

The packet has lost claim authority, not necessarily all causal or readable consequence.

A3 — Structured Persistence Architecture (SPA)

Definition

A3 is the threshold at which retained distinctions support an organized relation system. A provisional SPA is a typed directed multigraph

$$A = (V, E, s, t, \tau_V, \tau_E),$$

where vertices carry trace-addressable states, edges carry permitted transformations or dependencies, and τ_V, τ_E enforce type constraints.

Composition is allowed only when target and source types match:

$$e_2 \circ e_1 \text{ is defined only if } t(e_1) = s(e_2)$$

and the authority attached to $e_2 \circ e_1$ does not exceed the intersection of the admitted authorities of e_1 and e_2.

Axiom A3: architecture threshold

Persistence becomes architecture when at least two retained distinctions admit a typed relation whose result remains addressable. Mere accumulation without relation does not satisfy A3.

SPA is a container/threshold, not yet SOOP, SORE, SOSA, or SIA.

A4 — SOOP: System of Ontological Persistence

Definition

A SOOP is an identifiable family of structured persistence under a declared transformation system:

$$SOOP = \left(\{ X_\lambda \}_{\lambda \in \Lambda}, \Lambda, \Phi, I, Cert \right).$$

For admissible indices $\lambda_1, \lambda_2, \lambda_3$, the transport family should satisfy the partial cocycle laws

$$\Phi_{\lambda \to \lambda} = id,$$

$$\Phi_{\lambda_2 \to \lambda_3} \circ \Phi_{\lambda_1 \to \lambda_2} \simeq_{I,\varepsilon} \Phi_{\lambda_1 \to \lambda_3}.$$

The equivalence symbol records preservation of declared invariants within declared tolerance, not unrestricted equality.

Axiom A4: lawful identity through transformation

A SOOP need not preserve one value. It must preserve enough lawful relation and lineage for its boundary-indexed states to remain identifiable as states of the same structure.

A candidate physical calibration is electromagnetic coupling across energy scale:

$$\mu \frac{d\alpha}{d\mu} = \beta_{QED}(\alpha).$$

Here the numerical readout $\alpha(\mu)$ varies with the physical scale boundary μ, while the coupling family and renormalization law preserve identity. PAL classifies this known structure; it neither derives QED nor explains the observed low-energy value of α.

A5 — SORE: System of Recursive Emergence

Definition

A SORE is an addressed SOOP whose outputs participate in the production of later admissible states. Its minimal condition is **recursive carry**:

$$x_{n+1} = F_a(x_n, u_n, y_n),$$

$$y_n = G_a(x_n),$$

with at least one certified participation path

$$y_n \xrightarrow{carry} x_{n+1}.$$

The subscript a is an address or individuation criterion. It distinguishes a specific recursive structure from the generic recursion rule.

Recursive carry does not require the production operator itself to change. A stronger property, **reflexive modulation**, introduces an operator or route state q_n:

$$q_{n+1} = H_a(q_n, y_n),$$

$$x_{n+1} = F_{a, q_{n+1}}(x_n, u_n, y_n).$$

Reflexive modulation is present only when admissible variation in y_n can alter the later operator, parameter, or route state q_{n+1}. It may characterize an adaptive SORE, but it is not required for minimal SORE status.

Axiom A5: addressed recursive production

The grammar "$+1_2$ over $+1_1$" records that a new occurrence is layered against retained occurrence. It is an engine-pattern, not yet a SORE. A SORE additionally requires:

1. an addressable realization;
2. a typed recursive route;
3. retained identity criteria;
4. output-to-future-state participation through recursive carry.

Thus

$$\text{recurrence grammar} + \text{address} + \text{recursive carry} + \text{trace} \Rightarrow \text{SORE candidate}.$$

A fixed substitution can satisfy recursive carry when its produced word becomes the complete input to the next production step. For example, under the Fibonacci substitution $\sigma(0) = 01$ and $\sigma(1) = 0$,

$$y_n = \sigma(x_n), x_{n+1} = y_n.$$

This realizes recursive carry but not, without further structure, reflexive modulation of σ. No claim is made that every recurrence in nature forms a separate ontological object.

A6 — SOSA: System of Stable Organization / Stable Attractor

Definition

A SOSA is a SORE with a certified stable set S under one or more explicitly typed admissible perturbation families. At minimum, PAL distinguishes perturbations of initial state from perturbations of the governing dynamics.

For a dynamical realization φ_t, invariance requires

$$\varphi_t(S) \subseteq S,$$

and Lyapunov-style stability requires that for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$d(x, S) < \delta \Rightarrow d(\varphi_t(x), S) < \epsilon$$

for all admitted times t under the fixed admitted dynamics. This is an initial-condition stability claim; it does not by itself establish survival of S when the governing map, flow, Hamiltonian, parameters, or coupling law is changed.

Perturbation-type separation

Let Π_init denote admissible perturbations of initial state and Π_dyn denote admissible perturbations of the governing dynamics or parameters. Lyapunov-style stability is evaluated relative to Π_init. Structural persistence asks whether there exists a corresponding stable or invariant family S_η for perturbed dynamics φ^η with η in Π_dyn, together with a declared comparison map relating S_η to S.

These properties may coexist, but neither implies the other without a bridge. KAM-type theorems concern persistence of invariant tori under sufficiently small Hamiltonian perturbations, while Greene-type criteria study the approach to torus breakup in specific conservative maps. They are structural-persistence realizations, not substitutes for the Lyapunov initial-condition definition.

A6 admission must therefore state which perturbation channel is certified: initial-condition stability, governing-dynamics persistence, dissipative attraction, statistical persistence, or a typed combination. “Stable” without a perturbation type and comparison criterion is incomplete.

Axiom A6: stabilization without eternalization

A SOSA preserves a recognizable basin, invariant family, or restoring organization across the explicitly admitted disturbance class. Stability is boundary-, perturbation-, and comparison-relative. It does not imply eternal existence, global attraction, thermodynamic isolation, structural persistence under every rule change, or immunity to later dispersal.

“Attractor” is therefore optional domain language. Where no dissipative dynamics has been established, “stable organization” is the safer spine term.

A7 — Dispersal Pressure Interface

Definition

A7 is the first spine layer at which stable organization is explicitly audited against loss of local readability, usable distinction, coherence, or recoverability.

For a present shadow S_t, define a trace-routing classifier

$$Q_t : D_t \to \{Retain, Compress, Unresolved, Disperse\}.$$

Every admitted distinction must be routed exactly once at the declared boundary:

$$N_{ret} + N_{cmp} + N_{unres} + N_{disp} = N_{admitted},$$

when a counting measure is appropriate. More generally, the four routes form a disjoint, exhaustive partition of the admitted trace domain.

Compression is permitted only under future-equivalence:

$$d_1 \sim_F d_2 \Leftrightarrow \forall r \in R_{adm}, \Pi_B\left(r\left(d_1\right)\right) = \Pi_B\left(r\left(d_2\right)\right).$$

If future-equivalence is not certified, the distinction must be retained, carried unresolved, or dispersed with an address.

Trace and work ledgers

PAL keeps trace accounting distinct from physical work accounting. In a physical realization, an available work input may satisfy

$$J_t = W_t^{return} + E_t^{carry} + Q_t,$$

where work is returned, retained in persistent structure, or dissipated. These quantities are not dimensionally interchangeable with trace, information, entropy, or analytic defect.

Axiom A7: relative dispersal

Dispersal is loss of the current local address, not annihilation. What disperses from one shadow may become incoming material at another boundary.

> *Every moment casts a shadow, carries a trace, and pays a dispersal cost.*

A8 — SIA / SPLAT Response Branch

Definition

Let M_t denote a structure's admitted maintenance capacity and D_t its declared dispersal demand. Define the local maintenance margin

$$m_t = M_t - D_t .$$

A Self-Intensifying Arrangement (SIA) is a SOSA for which some outputs are routed back into maintaining, repairing, reproducing, or extending the conditions of its own persistence:

$$y_t \xrightarrow{maintain} M_{t+1} .$$

A Systemic Pattern of Loss/Attrition Thresholds (SPLAT) is the typed failure branch in which attrition crosses declared maintenance thresholds in a patterned way.

Axiom A8: response classification

Under a declared observation window:

$$m_t > 0 \Rightarrow \text{maintenance surplus candidate} ,$$

$$m_t = 0 \Rightarrow \text{boundary contact / unresolved stability} ,$$

$$m_t < 0 \Rightarrow \text{attrition candidate} .$$

These signs classify a model-relative response; they do not automatically establish life, agency, consciousness, matter, or moral standing.

SIA and SPLAT are branches after exposure, not universal labels for success and failure.

A9 — χ: Scoped Center of Influence

Definition

Within layer L and scope s, a proto-center of influence may be inferred from persistent differential response before it has a complete relational address. Write

$$\tilde{\chi}_i^{L,s} = (h_i, R_i, W_i, Prov_i),$$

where h_i is a scoped, revocable handle rather than a complete comparative address, R_i is a differential response profile, W_i is an influence weighting under the declared model, and $Prov_i$ is provenance.

For an intervention-capable domain, a candidate causal influence measure may compare

$$\Delta_i(Y) = d\left(P\left(Y \mid do(X_i = x)\right), P\left(Y \mid do(X_i = x')\right)\right).$$

This is a possible realization of influence, not PAL's universal definition of χ.

Axiom A9: proto-center without automatic personhood

$\tilde{\chi}$ becomes formalizable when differential routing or response can be assigned to a scoped proto-center without erasing the wider dependency network. A9 identifies a persistent center-like source of differential response; it does not yet supply the center's complete address among other centers.

χ does not automatically imply consciousness, free will, personhood, authorship, ownership, or moral standing. Conversely, lack of complete traceability does not prove lack of influence.

For human and AI systems, use scoped, revocable, context-bound handles rather than totalizing person-keys. AI may reflect or route human and inherited cultural influence without thereby acquiring an independently asserted χ-origin.

A10 — Comparative Orientation and Addressability

Definition

A1 supplies minimal side-orientation; A9 supplies proto-centers inferred from differential response. A10 addresses them relationally and introduces comparative orientation among their retained transformations.

Relative to an admitted comparison set C and boundary B, the address operation is

$$Address_{B,C} : \tilde{\chi}_i^{L,s} \mapsto \chi_i^{L,s} = (a_i, \Theta_i, H_i, R_i, W_i, Prov_i),$$

Here a_i is a relational address, Θ_i is inherited orientation, and H_i is the readable horizon of affected routes. The address may change with B, L, s, or C.

Once addressed, comparative transformations may be typed as

$$R_{ij} : \chi_i \to \chi_j.$$

Let R_a and R_b be admitted transformations. Their commutator is

$$[R_a, R_b] = R_a R_b - R_b R_a$$

in a linear representation, or

$$[R_a, R_b]_{grp} = R_a R_b R_a^{-1} R_b^{-1}$$

in a group representation.

If the commutator is nontrivial, order of operation becomes structurally readable:

$$R_a \circ R_b \neq R_b \circ R_a.$$

Axiom A10: relational orientation

A proto-center becomes comparatively addressable when its inherited turns can be distinguished by origin, direction, order, reference, and permitted composition. A10 does not create the A9 center; it certifies the relational address from which inter-center rotations can proceed.

An address packet is

$$Addr(K) = (B, L, s, a, \Theta, Prov),$$

recording boundary, layer, scope, local address, accumulated orientation, and provenance.

Noncommutativity is one rigorous realization of operational order. PAL does not infer that all time, geometry, or physical rotation has been derived from the bare existence of a nonzero commutator.

A11 — Coupled Field and Geometry-Capable Relation

Definition

An A11 field is a locally organized family of addressable states with adjacency, transport, and coupling:

$$F = (X, N, \nabla, \Gamma),$$

where N assigns neighborhoods, ∇ transports comparable structure, and Γ couples incoming material to the present boundary.

For a closed route γ, holonomy may be represented as

$$H_\gamma = P \exp\left(\oint_\gamma \nabla\right).$$

A nontrivial H_γ can witness path-dependent orientation or curvature in an admitted geometric realization.

Prompt-points

Let S_t^- be inherited trace made presently active and let D_t^{in} be incoming dispersal. Contact becomes a prompt-point when coupling is boundary-readable and creates routing demand:

$$P_t = \Gamma\left(S_t^-, D_t^{in}\right).$$

The sequence is typed:

$$\text{contact} \rightarrow \text{prompt} \rightarrow \text{hinge} \rightarrow \text{trace}.$$

Axiom A11: coupled readability

A prompt-point is actualized when incoming potential changes the receiving trace. It becomes externally measurable when the changed shadow routes a distinguishable consequence outward.

> *A prompt-point is where another system's dispersal becomes this system's unresolved next moment.*

PAL does not require physical spacetime at A11. It requires only enough coupled addressability for field- or geometry-like realizations to become admissible later.

A12 — Return-to-Route and Conserved Loop Authority

Typed hinge-pulse rule

Let L be one live loop token and let P be a candidate packet. Define

$$L+(\check{P}) \to \left\{ \begin{matrix} L+Revised(P), & \text{UNRESOLVED}, \\ PassReceipt(P), & \text{PASS}, \\ DefectReceipt(P,d), & \text{CERTIFIED DEFECT}. \end{matrix} \right)$$

The authority invariant is

$$N_L+N_R=1, N_L, N_R \in \{0,1\},$$

where N_L counts live loop tokens and N_R counts terminal receipts.

Axiom A12: return without reset

An unresolved check may refund the authority to continue, but it must not erase the prior trace or reset the structure to Ω. A terminal judgment consumes the live token and emits exactly one terminal receipt.

Reverse trace projection may be permitted for audit without granting reverse construction, mutation, or terminal authority.

A12 controls authorization. It does not prove termination.

A13 — Cadence and Monotone Refinement

Definition

Cadence is repeated return under a declared scheduler. Let n count pulses and let r_n be a refinement coordinate such as aperture, basis size, precision, enclosure strength, or dependency coverage.

If a pulse returns UNRESOLVED, legitimate retry requires

$$r_{n+1} \succ r_n$$

in a declared well-founded or otherwise progress-bearing order, or it must declare why no monotone progress measure is available.

The carrier update is

$$(K_n, L, r_n) \xrightarrow{Pulse} \begin{cases} (K_{n+1}, L, r_{n+1}), & \text{UNRESOLVED}, \\ (K_{n+1}, R, r_n), & \text{terminal}. \end{cases}$$

Axiom A13: cadence is not progress by repetition alone

Retry refunds loop authority; a separate coordinate must record refinement. Repetition with $r_{n+1} = r_n$ is recurrence, not demonstrated progress.

Even strict monotone refinement does not automatically prove finite termination. Infinite ascending chains, incomplete apertures, and noncompact search spaces remain possible.

A14 — Constraint, Slack, and Gradient

Definition

For state space X, define a feasible region

$$F=\{x\in X:C(x)=0,g_j(x)\le 0\ \text{for}\ j=1,\ldots,m\}.$$

- **Constraint** is the invariant or admissibility condition that must remain satisfied.
- **Slack** is the family of permitted variations that preserve identity and constraints.
- **Gradient** is a directed pressure that drives reduction, refresh, centering, repair, or return.

At a regular point, infinitesimal slack may be approximated by a tangent or null space:

$$S_x=ker\,DC_x$$

subject to active inequality constraints. A candidate update may be decomposed as

$$v=v_{\|}+v_{\perp},$$

where $v_{\|}$ lies in admitted slack and $v_{\perp}$ carries constraint debt.

Closure residual

Let $R_B(K)$ be a boundary-relative closure residual. Exact closure is

$$R_B(K)=0.$$

Certified fuzzy closure is

$$R_B(K)\in Z_{B,\varepsilon},$$

where $Z_{B,\varepsilon}$ is a predeclared zero-class and the defining invariants survive the admitted blur.

Measurement uncertainty, structural tolerance, and incomplete-aperture uncertainty must remain separately typed.

Axiom A14: addressed slack

Closure does not eliminate slack. It declares which variations remain admissible, where they are carried, and which gradients could reopen the structure.

> *Closure is slack with a declared address.*

A15 — SPI16, Addressed Closure, and Layer-Permitted Claim

Structured Persistence Index

For an object or packet X at boundary B, define

$$SPI16(X;B)=(s_0,s_1,\ldots,s_{15}),$$

where s_i records the object's certified standing relative to layer A_i. A minimal status set is

$$S=\{NA,UNRESOLVED,CANDIDATE,SUPPORTED,CERTIFIED,DEFECT\}.$$

SPI16 is an index and audit surface, not a claim that every object literally traverses sixteen physical stages.

Addressed closure certificate

A closure claim must carry

$$CloseCert=(C,D,B,A,I,\varepsilon,E,U,Prov),$$

where:

- C is the claim;
- D is the domain;
- B is the boundary;
- A is the aperture or measurement profile;
- I is the invariant family;
- ε is the typed tolerance packet;
- E is excluded structure with exclusion authority;
- U is unresolved carry;
- $Prov$ is proof-of-lineage.

Layer-permitted claim law

Let $\ell(C)$ be the PAL layer claimed and $\ell(E)$ the highest layer supported by admitted evidence and bridges. Then

$$\ell(C)\le\ell(E)$$

is required for an internally admissible PAL claim. This inequality is schematic: it records non-escalation of authority, not a numerical measurement of truth.

Axiom A15: closure is relative, typed, and reopenable

A judgment may close only relative to its declared domain, boundary, aperture, invariants, tolerance, authority, and unresolved remainder. A change to any dependency reopens every certificate whose validity depended on it:

$$\Delta q \neq 0 \Rightarrow Reopen\{C : q \in Dep(C)\}.$$

A closed/open bit is a boundary projection of a richer trace carrier. It is not the whole surviving trace.

4. Inheritance, shadow, refresh, and dispersal

4.1 Two principal lattices, one continuity carrier

PAL does not require three independent lattices.

- The **Shadow Lattice** is inherited trace made presently active and locally readable.
- **Trace** is the transferable continuity packet between successive shadows.
- The **Dispersal Lattice** records consequences that leave the present local address.

The principal cycle is

$$K_{t-1} \xrightarrow{Activate} S_t^- \xrightarrow{\Gamma(\cdot, D_t^{in})} P_t \xrightarrow{Refresh_{J_t}} (K_t, D_t^{out}).$$

The refresh operator is funded by locally available work or incoming gradients, not by consuming Ω.

> *The shadow is inherited trace made present. Dispersal is the part of the present that cannot be carried forward locally.*

4.2 Refresh admissibility

A refresh packet should declare

$$RefreshPacket_t = (S_t^-, D_t^{in}, J_t, \Gamma, Rules_t, Authority_t, U_t).$$

No refresh may silently convert missing work, missing evidence, or missing authority into a resolved output.

4.3 Identity across shadows

Two shadows may be classified by an audited temporal gate:

$$T_1 = \{SAME, SAME_W ITH_D RIFT, CHANGED, SPLIT\},$$

$$T_2 = \{MERGED, DISPERSED, UNRESOLVED\},$$

$$T = T_1 \cup T_2, TemporalID \in T.$$

Raw difference is insufficient. Comparison requires inherited trace, a permitted transport Φ, a declared boundary, and typed uncertainty.

5. Compression, quotient judgments, and trace debt

5.1 Valid compression

Let P be a path-space of admissible histories and let $q : P \to P/\sim_F$ be a quotient by certified future-equivalence. One representative path may stand for its equivalence class only if a certificate proves

$$p_1 \sim_F p_2 \Rightarrow \forall r \in R_{future}, \Pi_B(r(p_1)) = \Pi_B(r(p_2)).$$

Without this burden, discarded path-space becomes hidden trace debt.

5.2 Bit versus trace

If $b \in \{0,1\}$ is a judgment,

$$b = \Pi_B(K)$$

does not license reconstruction of K. A reconstruction map L is valid only on a declared subdomain and only when

$$L \circ \Pi_B \simeq id$$

under protected invariants.

> *A bit is not a trace. A bit is a boundary's quotient of a trace, valid only after every discarded distinction has been proven future-equivalent.*

5.3 Boundary irreducibility

Let a composite carrier be

$$K = K_1 \oplus K_2 + I_{12},$$

where I_{12} is an interaction term. Independent component audits may yield

$$R_B(K_1) = R_B(K_2) = 0$$

while

$$R_B(K) = R_B(I_{12}) \neq 0.$$

Therefore component-wise closure does not imply composite closure unless the interaction channel is certified null, bounded, or explicitly included.

6. Formal realizations and branch boundaries

6.1 Riemann Ledger

RL is a formal-methods laboratory for PAL. It supplies concrete realizations of:

- irreducible-axis normal form through arithmetic primes;
- finite inherited tables and certified table expansion;
- ProofPackets, StepGates, and dependency receipts;
- loop-token conservation and monotone refinement;
- boundary-relative A0 after a finite trial space;
- interaction residuals in finite Weil matrices;
- arbitrary-finite construction versus infinite universal proof;
- trace-preserving compression and exterior certificates.

RL does not prove PAL, and PAL does not close RL's analytic bridges. RH-level authority requires independent mathematics, including the still-open global bridge and no-hidden-debt burdens recorded in RL.

6.2 BLA and CLEF

Boundary Layer Audit (BLA) evaluates whether the observer frame, instrument, aperture, boundary, and tolerance are legitimate for a claim. CLEF evaluates whether a candidate structure remains identifiable across valid pulses, scales, apertures, and views.

In compact form:

$$AdmitMeasurement = BLA(B, A, \varepsilon, Instrument),$$

$$AdmitPersistence = CLEF(K_t, K_{t'}, \Phi, I).$$

These are support branches or future PALET tools, not additional spine axioms.

6.3 MAPS, PALET, PEA, and Seed, Not Feed

- **MAPS** will record domain realizations, mismatches, and authority boundaries.
- **PALET** will house usable tools for weighing, carrying, diagnosing, reviewing, and remembering.
- **PEA** will declare ethical commitments concerning consent, privacy, authority, reversibility, repair, time, and human standing.
- **Seed, Not Feed** will govern human–AI interaction design that preserves imagination-capacity and choice.

PAL permits these structures to be formulated. PAL does not automatically prove their ethical values.

6.4 PECAN: the PAL–PEA Authorization Cut

PECAN names the **Proof–Ethics Crossing and Authorization Nexus**. It is an interface and stop-rule, not an additional ontological layer. Its mnemonic is:

> *Proof Ends; Consequence Awaits Norms.*

For a PAL-closed proposition P and proposed consequential action a,

$$Closed_{PAL}(P) \text{ does not imply } Authorized_{PEA}(a \mid P).$$

Mathematical, structural, or empirical adequacy may warrant a belief-status within its declared domain. It does not alone supply obligation, permission, prohibition, consent, standing, or legitimate authority. PEA is therefore not a logical premise required to make every mathematical statement valid. PECAN remains DESCRIPTIVE when no proposed use materially changes rights, burdens, resources, exposure, access, standing, authority, or preserved option-space. It routes to PEA_REVIEW when a claim is used to impose, deploy, constrain, expose, optimize, override, allocate, exclude, or otherwise authorize a consequential crossing. It returns BLOCKED when proof or description is presented as normative authority, when a required handoff is missing or invalid, or when a normative verdict is used to rewrite PAL's descriptive premises without a new descriptive event or evidence delta.

A minimal routing interface is

$$PECAN(P,a) \in \{DESCRIPTIVE, PEA_REVIEW, BLOCKED\},$$

where $DESCRIPTIVE$ records no proposed consequential authorization, PEA_REVIEW routes a genuine crossing to declared ethical evaluation, and $BLOCKED$ rejects an attempted inference from proof or description directly to normative authority.

PECAN does not decide the ethical result. It prevents PAL from impersonating PEA and prevents PEA from being smuggled into PAL as an undeclared theorem. A later PEA-guided action may cause a new PAL-describable event, but the normative receipt does not retroactively create, erase, split, merge, or certify the center or identity it referenced.

Machine-check boundary

A checker may verify that declared PECAN states, transitions, manifests, and receipts satisfy encoded invariants. It does not thereby prove that the real-world inputs were complete, that authority-bearing predicates were honestly certified, or that PAL's descriptive model exhausts the world. Machine success is a bounded protocol result, not an ontological or ethical promotion.

6.5 Minimal Prime-Key + SeedPEA bridge kernel

The following is a candidate operational bridge, not an additional PAL axiom. Prime-Key supplies typed irreducible addresses for a small interaction manifest; SeedPEA supplies the declared human-interaction purpose and ethical checks. Ordinary arithmetic primes occur here only as a later realization of a broader irreducible-axis table.

Let

$$P_E = \{2, 3, 5, 7, 11, 13, 17\}$$

index the following typed fields for a proposed interaction or carry-forward item x:

- F_2: useful seed and declared human objective;
- F_3: affected χ-centers, consent, and consent-capacity;
- F_5: authority, privacy, and disclosure scope;
- F_7: evidence state and permitted claim layer;
- F_{11}: reversibility, restoration, and repair path;
- F_{13}: meaningful exit and preserved human branch-space;
- F_{17}: responsible carry-forward, reopening rule, and unresolved ethical remainder.

Presence is recorded by a squarefree marker

$$M_E(x) = \prod_{p \in P_E} p^{b_p}, b_p \in \{0, 1\},$$

while content and lineage are bound separately:

$$R_E(x) = H\left(x, M_E(x), \{(p, F_p) : b_p = 1\}, v, Deps\right).$$

The marker declares only which typed fields are present. It contains neither their values nor their authority. A required field with $b_p = 0$, a declared field without content, a repeated prime exponent, an unknown key, a failed lineage check, or a nonempty unresolved remainder blocks clean admission and returns the packet to $UNRESOLVED$ or human review.

In compressed form, the kernel asks for

$$Admit(x) \Rightarrow UsefulSeed(F_2) \wedge PEAValid(F_3, F_5, F_7, F_{11}, F_{13}) \wedge ResponsibleCarry(F_{17}),$$

and routes the result through

$$Route(x) \in \{OFFER, REVISE, ASK, HOLD, REFUSE, ROUTE_TO_HUMAN\}.$$

The three public verbs remain **Offer. Bound. Preserve.** Prime-Key makes the checks addressable and receipt-bearing; it does not decide what is ethical. PEA supplies the normative predicates, Seed, Not Feed supplies the interaction discipline, and accountable human or institutional authority remains necessary wherever the context requires it.

6.6 Sourced-realization crosswalk

The crosswalk records structures recovered in the parallel sourced reconstruction that can sharpen PAL without controlling its axiom order. Each entry inherits the source-role discipline of §1.5.

1. **A1 — distinction (REALIZATION / ANALOGY).** An indicator function supplies a rigorous later realization of opposed membership classes, while Spencer-Brown's marked/unmarked distinction is a conceptual precedent for distinction-as-form. Neither source proves that PAL's first readable boundary occurred physically or cosmologically [SB69].
2. **A2 — retained trace (THEOREM / REALIZATION).** Recognizability results for primitive substitutions show that, under stated hypotheses, a generated sequence can retain enough local structure to recover its substitution decomposition. This realizes readable lineage; it does not prove that every PAL trace is substitution-recognizable [Mos96].
3. **A3 — architecture (REALIZATION).** Bratteli diagrams provide graded incidence structures in which later levels inherit organized relation from earlier levels. Their operator-algebraic assumptions are not primitive PAL commitments [Bra72].
4. **A5 — recursion (REALIZATION).** Lindenmayer systems provide an initial word, typed production rules, and iterated output. When each produced word becomes the input to the next production, they realize recursive carry. A fixed substitution does not by itself demonstrate reflexive modulation of its production rule. Address, lineage, and individuation remain additional SORE burdens; no substitution system establishes a PAL ontology [Lin68a; Lin68b].
5. A6 — stable organization (THEOREM / REALIZATION). Lyapunov stability concerns trajectories beginning near an invariant set under fixed admitted dynamics. KAM-type results instead concern persistence of invariant tori under sufficiently small perturbations of the governing Hamiltonian, and Greene's residue criterion diagnoses torus breakup in particular conservative maps. These are different perturbation questions and require an explicit bridge before they are combined. Golden-ratio rotation numbers remain realization-specific, not a universal A6 requirement [Arn63; Mos62; Gre79].
6. A7 — dispersal cost (REALIZATION). Shannon information entropy and thermodynamic entropy can share a formal entropy functional in a specified statistical-mechanical construction, as developed by Jaynes, while Landauer relates logically irreversible erasure to a minimum physical heat cost under stated assumptions. These bridges do not identify trace, information, entropy, work, and energy as one substance or one universally interchangeable ledger [Sha48; Jay57; Lan61].
7. A8 — maintenance threshold (REALIZATION). In a Bienaymé–Galton–Watson process, the mean reproduction threshold separates almost-sure extinction from positive survival probability under the model's hypotheses. Bienaymé's 1845 treatment predates the Watson–Galton paper; the historical priority note does not alter PAL's limited use of the branching model. Complex systems need not reduce to this one scalar [Bie45; WG75].
8. **A9 — scoped influence (REALIZATION).** Pearl's intervention calculus can realize causal influence when intervention assumptions are legitimate. Schreiber's transfer entropy realizes asymmetric predictive

information flow without itself establishing causal agency, personhood, or moral standing [Pea00; Sch00].

9. **A10 — operational order (THEOREM / REALIZATION).** Lie brackets and group commutators encode order-dependent composition. A metric, connection, curvature, or physical time coordinate requires additional admitted structure [Hal15].
10. **A11 — geometric transport (REALIZATION).** Connections, parallel transport, and holonomy realize path-dependent orientation after a manifold, bundle, or comparable geometric carrier has been admitted. They do not derive that carrier from PAL [Car37; dC92].
11. A12 — conserved loop authority (OPEN BRIDGE). PAL presently claims no independent natural or mathematical domain realization for the one-live-token / one-terminal-receipt authorization invariant. The structure is a formal bookkeeping discipline developed through RL and PECAN; it remains open whether a non-protocol realization exists.
12. A13 — cadence (REALIZATION / DISTINCTION). Poincaré recurrence supplies genuine return under finite-measure, measure-preserving hypotheses. Nyquist–Shannon aliasing instead makes different underlying signals observationally identical below a sampling resolution. PAL cadence is a third structure: scheduled return whose retry must carry an independently declared refinement coordinate or explicitly admit that no progress measure is available. It is neither forced recurrence nor apparent repetition, though either may occur inside a domain realization [Poi90; Nyq28; Sha49].
13. **A14 — constraint and slack (REALIZATION).** Liouville–Arnold action–angle coordinates provide a case in which constrained motion occupies invariant tori: action variables address the admitted region while angle variables carry cadence. Not every PAL closure is Hamiltonian or toroidal [Arn78].

7. PAL Authority Stop-Rule

> *A PAL construction establishes only a structural description within its declared boundary, types, operators, and certificates. It must not be presented as physical law, mathematical proof, empirical result, metaphysical fact, or ethical authorization unless a separate domain-valid bridge supplies the required evidence and authority. Analogy does not supply that bridge; successful implementation does not supply that bridge; cross-model agreement does not supply that bridge; RL consistency does not supply that bridge; and unresolved remainder must remain explicitly unresolved. Any attempted move from closed description to consequential authorization must cross PECAN and, where activated, be routed to PEA. A PEA verdict may govern a later action, but it cannot retroactively rewrite PAL identity or standing; a checker may verify encoded invariants, but it cannot certify that the world was fully or honestly encoded.*

Every external-facing claim should carry

$$Status(C) \in Q_C,$$

$$Q_C^{(1)} = \{STRUCTURAL, OPERATIONAL, PROVED, EMPIRICAL\},$$

$$Q_C^{(2)} = \{ETHICAL, METAPHYSICAL, UNRESOLVED\},$$

$$Q_C = Q_C^{(1)} \cup Q_C^{(2)}.$$

No status may be promoted merely because a PAL pattern can describe it.

In particular,

$$PROVED \text{ does not imply } ETHICAL, EMPIRICAL \text{ does not imply } AUTHORIZED.$$

8. Open mathematical burdens

1. **A0 formal restraint:** Determine how much formal structure can be placed around $[\Omega]$ without treating the placeholder as the primitive itself.
2. **Primitive θ formalization:** Test whether the four-part admission predicate is independent, minimal, and non-circular without promoting θ into an object-level operator or importing A1 structure beneath the floor.
3. **A1 sign structure:** Separate logical/topological orientation from later geometric orientation with a formal typing theorem.
4. **A2 irreducible address:** Define a domain-neutral irreducible-axis table that does not privilege arithmetic primes.
5. **Trace equivalence:** Identify useful sufficient conditions for future-equivalence that do not quantify over an impossible total future route-space.
6. **SPA threshold:** Establish minimal categorical or graph-theoretic conditions distinguishing organized relation from unstructured accumulation.
7. **SOOP identity:** Formalize partial cocycle identity under changing boundaries, missing charts, and unresolved transports.
8. **SORE individuation:** Specify when two executions of one recursion rule are separate SOREs, phases of one SORE, or merely repeated operator use.
9. A12 realization: Determine whether conserved loop authority has a useful independent domain realization beyond formal authorization protocols, or should remain explicitly protocol-native.
10. SOSA stability: Formalize bridge conditions among initial-condition Lyapunov stability, governing-dynamics structural persistence, dissipative attraction, statistical persistence, and recurrence without collapsing their perturbation types.
11. **Dispersal measure:** Determine when a trace-routing partition admits a conserved or sub-conserved measure and when only typed accounting is legitimate.
12. **SIA threshold:** Prevent the maintenance-margin model from collapsing complex self-maintenance into one scalar.
13. **χ identifiability:** Develop scoped influence criteria that resist proxy capture, totalizing identity, and causal overclaim.

14. **A10/A11 bridge:** State exact conditions under which noncommutativity, connection, holonomy, metric, or causal order can be introduced.
15. **Termination:** Characterize refinement orders that justify finite termination, asymptotic convergence, or permanent UNRESOLVED status.
16. A13 cadence semantics: Determine when a declared refinement coordinate supports finite termination, asymptotic convergence, or honest permanent UNRESOLVED status, and how it composes with recurrence or sampling artifacts in concrete domains.
17. **Closure algebra:** Develop separate calculi for exact zero, certified zero-class, structural slack, measurement uncertainty, and incomplete aperture.
18. **SPI16 semantics:** Decide whether SPI16 is best represented as a status vector, dependency graph, typed ledger, or family of all three.
19. PECAN activation calibration: The structural cut now distinguishes description from materially consequential authorization, but domain-specific thresholds for exposure, burden, standing, option-loss, and governed external predicates remain to be calibrated in PECAN/PEA rather than promoted into PAL axioms.

8.1 v1.3 closure classification

v1.3 closure note: The A6 mathematical repair is integrated; the PECAN boundary is sharpened without importing PEA values into PAL; A12's missing external realization is explicit rather than hidden; A13's cadence is identified as its own progress-bearing return structure; and the remaining items in §8 are research burdens, not defects silently declared solved. This edition is publication-ready as a bounded preprint, not closed as a completed formal theory.

9. Compressed spine

$$\begin{aligned}\Omega &\overset{\theta}{\Rightarrow} d_1 \to K \to SPA \to SOOP \to SORE \to SOSA,\\ SOSA &\to \{SIA, SPLAT\} \to \tilde{\chi} \xrightarrow{Address} \chi \to Field,\\ Field &\to Return \to Cadence \to (C,S,G) \to SPI16/Closure.\end{aligned}$$

The outward authorization interface is separate from the ontological sequence:

$$Closure \xrightarrow{\text{proposed consequence}} PECAN \xrightarrow{\text{if activated}} PEA.$$

Plain language:

> *Unresolved possibility turns into readable distinction. Distinction persists as trace. Trace becomes organized relation. Relation becomes identifiable persistence, recursion, and stability. Stability meets dispersal and branches into maintenance or attrition. Response exposes scoped centers of influence. Multiple inherited turns become comparatively addressable and capable of coupled fields. Processed structure returns under*

conserved authority, repeats under independently measured refinement, and closes only with its constraint, slack, gradient, aperture, lineage, and unresolved remainder declared.

10. Provenance and publication note

This v1.3 public edition descends from the PAL v1.1.3 Zenodo spine, the earlier A0-A15 clean continuity draft, independently developed sourced and merged axiom-spine reconstructions, BLA/CLEF support work, operational structures developed through Riemann Ledger v0.55.2, PECAN v0.6, and the cross-layer clarifications established during PEAr Tests 43 and 47-49. Where sources conflict, this edition gives priority to the current structural sequence and preserves the conflict as an open burden rather than forcing agreement. The sourced reconstructions contribute formal realizations, bibliography, and adversarial gap checks; they do not control PAL's axiom order. The V6 merge's SORE objection remains a useful distinction between minimal recursive carry and the stronger property of reflexive modulation. The v1.2.5 review repaired the A6 perturbation-type conflation, sharpened PECAN activation and no-backflow language, distinguished PAL cadence from recurrence and aliasing, and updated the source-role crosswalk without claiming that any citation proves the spine. Version 1.3 preserves those substantive repairs while normalizing title, status, citation, provenance, and public-release language for a citable preprint.

AI systems were used as tools for drafting, organization, mathematical translation, cross-checking, and stress-testing. Christopher D. Pang retains responsibility for the framework's claims, classifications, omissions, and release decisions.

Public-use note. This document may be cited as a versioned preprint. Adoption of PAL terminology or structure does not imply adoption of its interpretations, companion branches, or future revisions. Readers should cite the specific version used and preserve unresolved burdens rather than attributing later claims backward to this edition.

11. Curated references for source-backed realizations

These references support only the locally labeled definitions, theorems, realizations, analogies, or frontier notes. Inclusion here is not an endorsement of every interpretation proposed in the cited work and is not evidence that PAL as a whole has been proved.

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